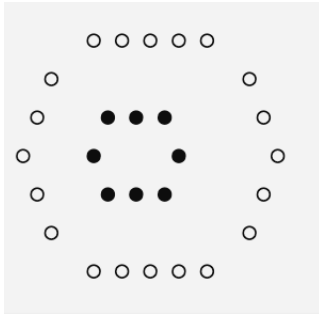


1. Suppose you have data that forms two concentric circles:



Would k-means be a good choice here? Why/why not?

2. Suppose you have 100 customers clustered around three locations:



X

The X is an extreme outlier. Will this be a problem for k-means? Why/ why not?

3. Consider:

- Cluster A: very dense
- Cluster B: very spread out

Both contain approximately the same number of points.

Why might K-means struggle?

4. True or false?

"If K-means converges, it has found the globally optimal clustering."

5. Suppose k-means produces the following results:

Initialization	Inertia
1	420
2	390
3	515
4	391

What does this suggest? What is your potential solution to this?

6. A hospital has patient records where each patient is already labeled as:

- High risk
- Medium risk
- Low risk

The hospital wants to predict the risk category for a new patient.

Should they use K-means?

7. Consider the data points:

$$A = (1, 1)$$

$$B = (2, 1)$$

$$C = (5, 4)$$

$$D = (6, 5)$$

Let $k = 2$

Initial centroids are:

$$\mu_1 = (1, 1)$$

$$\mu_2 = (6, 5)$$

Assign each point to its closest centroid and calculate the new centroids.

8. Suppose we have four customers:

Customer	Annual spending (\$1000s)	Purchases
A	2	1
B	3	2
C	8	7
D	9	8

Suppose $k = 2$

Initial centroids are:

$$\mu_1 = (2,1), \mu_2 = (2,1)$$

Perform one k-means iteration and interpret the resulting clusters